The Media Connectivity Association



OVERVIEW AND UPDATES · IMMERSIVE MEDIA

XR Media & 3D Scenes

Composing, positioning, and temporally synchronising dynamic 3D visual and audio objects for XR

MPEG (ISO/IEC) Khronos

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About the project

Composing, positioning, and temporally synchronising dynamic 3D visual and audio objects for XR

ISO/IEC 23090-14

MPEG standard

MPEG-I Scene Description: extensions to glTF 2.0 for timed media in a scene. Second edition 2025-11; Amendment 1 (FDAM 1, 2026-05-01) is at final draft.

ISO/IEC 12113

Khronos glTF 2.0

An open interoperable 3D asset format with a JSON scene description, which the MPEG-I Scene Description extensions build on.

3GPP TS 26.119

Technical Specification

Device media capabilities for AR services (MeCAR, created in Release 18), V19.1.0: device categories and the capabilities each supports, aligned with the OpenXR runtime API.

FIND OUT MORE **Technology** **Specifications** **Reference Tools**

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The scene description: ISO/IEC 23090-14

Overview and architecture of ISO/IEC 23090-14



Timed media in a scene

Features that describe how to get the timed media, and how a rendering process expects the data once it is decoded.

Extensions to glTF

The features are defined as extensions to the glTF format defined in ISO/IEC 12113.

Two interfaces

The MAF API and the Buffer API decouple the Presentation Engine from the Media Access Function.

Dynamic scenes

The scene description itself may change over time; the document defines how a change is signalled.

Carriage

Storage and delivery consistent with MPEG formats, using ISO/IEC 14496-12 and its derived specifications.

Advanced features

AR anchoring, interactivity, avatars and lighting; haptics and V3C objects.

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Features

The extensions of ISO/IEC 23090-14 to glTF 2.0

Timed framework external and timed media

MPEG_media | MPEG_accessor_timed | MPEG_buffer_circular

Dynamic and temporal media media and timing

MPEG_texture_video | MPEG_audio_spatial | MPEG_mesh_linking | MPEG_scene_dynamic | MPEG_viewport_recommended | MPEG_animation_timing

Real-time immersive and interactive media real-world and interactive scenes

MPEG_anchor | MPEG_scene_interactivity | MPEG_node_interactivity | MPEG_node_avatar | MPEG_lights_texture_based | MPEG_light_punctual | MPEG_haptic | MPEG_primitive_V3C | MPEG_sampler_YCbCr

Declare the source → **Bind into the scene** → **Timed accessors, circular buffers**

Timed media reaches the renderer in three steps; the Presentation Engine never needs to know the transport or codec of a media object.

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Architecture

Scene description reference architecture of ISO/IEC 23090-14

Receive	Request	Fetch and decode	Exchange
The Presentation Engine receives and parses the scene description document and its updates.	It identifies the timed media and requests it through the MAF API: where, which parts, when and in which format.	The Media Access Function instantiates the media fetching and decoding pipeline for the requested media.	The data is exchanged through buffers, controlled through the Buffer API.

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Repositories

On main: media through circular buffers, timed accessors and video textures; AR anchoring and interactivity in part

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-xr-unity-player	an XR-capable glTF scene viewer in Unity3D for the MPEG-I Scene Description extensions	Presentation Engine, Extensions	v1.1.0 27 Feb 2025	GitHub
rt-xr-glTFast	a glTFast fork that parses the MPEG-I Scene Description extensions	Extensions	none yet	GitHub
rt-xr-maf-native	the Media Access Function API, its C# bindings and media pipeline plugins	Media Access Function	none yet	GitHub
rt-xr-maf-plugin	a Unity3D package of the build artifacts of rt-xr-maf-native	Media Access Function	none yet	GitHub
rt-xr-content	reference scenes and media for testing and demos with the XR Player	Reference scenes	none yet	GitHub
rt-xr-blender-exporter	a Blender add-on that adds the MPEG_* extensions to the glTF exporter	Authoring	v1.1.0 16 Apr 2025	GitHub

FIND OUT MORE Technology Specifications Reference Tools

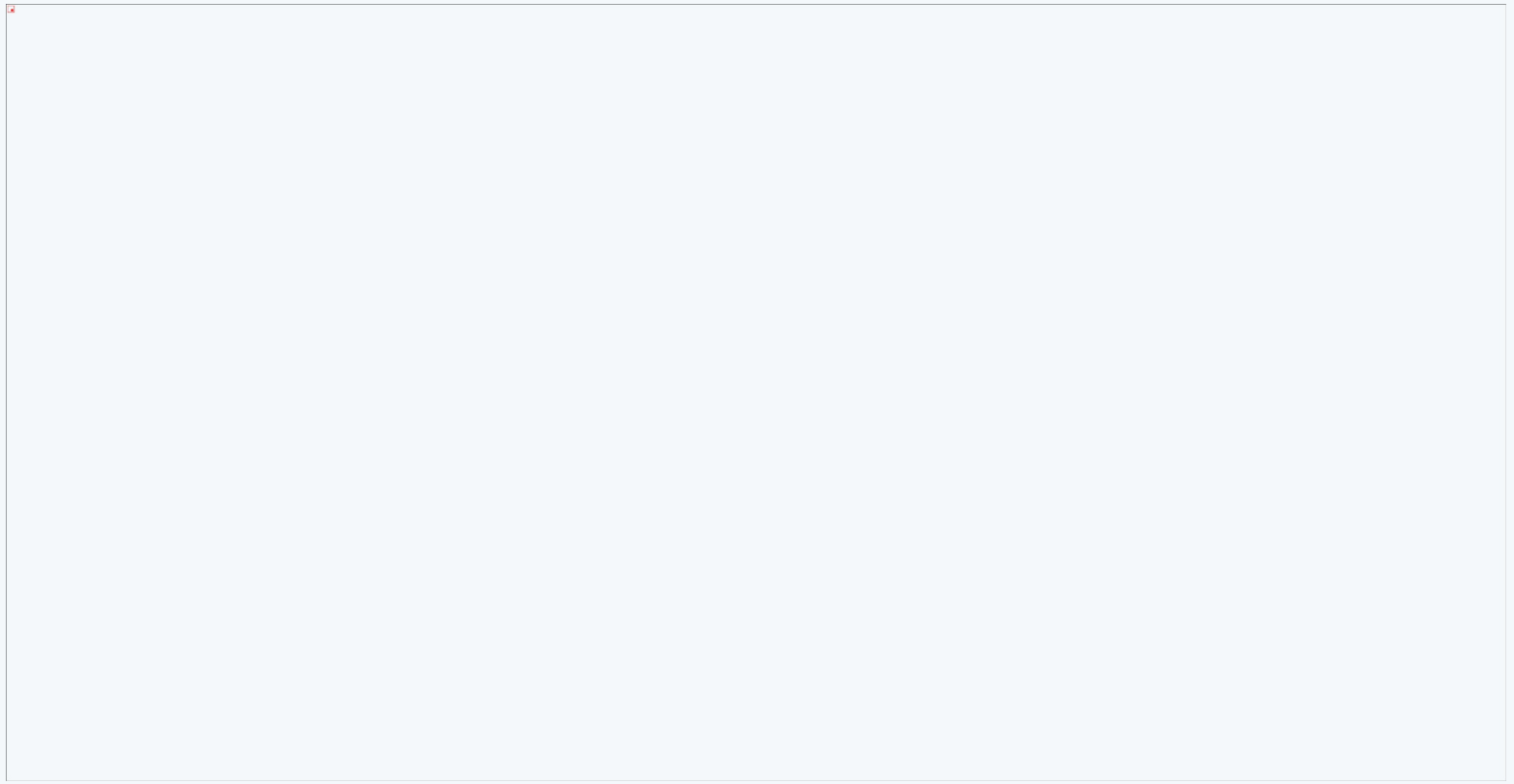
● released ● no release yet

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

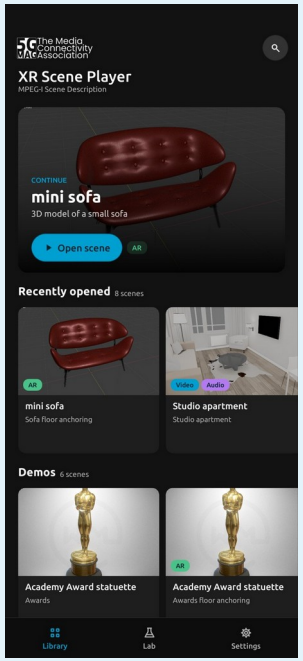
1. **rt-xr-maf-native**
2. **rt-xr-unity-player**
3. **rt-xr-glTFast**
   dependency, used by 2

Boxes without a badge are not mapped to a repository.

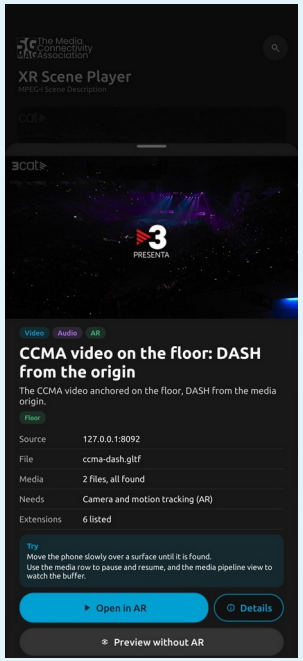
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See it and try it

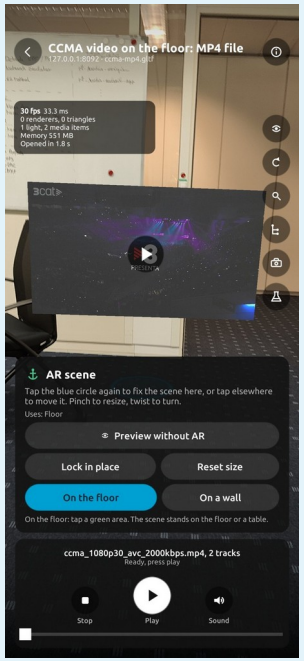
The XR Scene Player on an Android phone: pick a scene, open it in AR, control the media



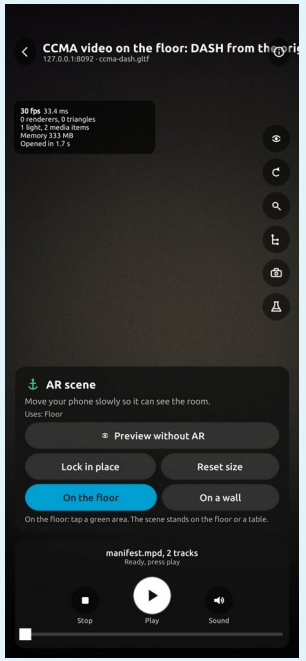
Library



Scene details



A video scene on the floor



AR and media controls

CONTRIBUTORS

MOTION SPELL interdigital. Fraunhofer FOKUS Qualcomm

WHERE WE STAND

XR Media & 3D Scenes

BROUGHT TO STUDY

- Support for new MPEG Extensions for glTF 2.0
- New XR Scene Player for Meta Quest
- Split rendering (TS 26.565, TS 26.249) and delivery of XR scenes over the RTC architecture

ON-GOING

- New XR Scene Player for Android
- Contributions to MPEG Scene validator
- Decoding of timed metadata tracks
- Support of media streaming of live content within XR/3D scenes

OUTCOMES

- Reference scenes aligned with ISO/IEC 23090-14
- Unity Player, Blender Project, Content repository

RELATION WITH SDOS AND INDUSTRY FORA

MPEG (ISO/IEC) Khronos

MPEG Contribution m78825 to ISO/IEC JTC 1/SC 29/WG 3 at MPEG#156, Hangzhou, October 2026

Khronos Initial inputs to the Khronos Exploratory Group on Remote Off-loading APIs

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